

Held-out Growing Source-Native Audits and Placement Sensitivity in the Twin-Prime Signed-Gram Diagnostic

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Abstract

We test whether the source-native finite V59 residual used in the preceding signed-Gram release survives a held-out origin/scale change, and whether its readout can be explained by source norms alone. The literal deleted-diagonal prime-shell operator is evaluated at two new origins, two larger scales, four shell anchors, two kernel exponents, and four fixed sign laws. The exact finite Gram identity separates the energy into a coordinate diagonal and an off-diagonal term. On the actual residual, the all-plus off-diagonal term is negative on 31/32 rows and positive on 1/32. A predeclared affine permutation preserves the source multiset and Euclidean L^2 norm, yet the all-plus term is positive on 32/32 permuted rows, with 31/32 classifications changed. The two-scale audit has 64 paired comparisons and all-plus energy growth factors in $[1.9663131482417533, 2.14326466572482]$. These are guarded finite numerical observations, not a growing arithmetic estimate. The principal contribution is a finite placement-sensitivity obstruction to reading the diagnostic from source norm or marginal values alone; no fixed-power credit, Route-B gate payment, or twin-prime conclusion is claimed.

1 Question and contribution

The current twin-prime route uses a literal centered prime-shell operator. A finite sign pattern can be misleading if it is tested at only one origin or if the operator is never exposed to the arithmetic source. The preceding release attached the finite V59 residual to the operator. We now make the smallest natural continuation: hold the operator and source model fixed, move to two previously unused origins and larger windows, and add a hostile control that preserves the source values as a multiset.

The paper contributes four finite facts. First, it provides a two-origin, two-scale holdout with an explicit scale-pairing audit. Second, it proves the same exact source-coordinate Gram decomposition for every finite row. Third, it applies the fixed affine map

$$\pi(i) = (5i + 17) \pmod{M}, \quad M = N/2,$$

to the source coordinates. Finally, it records the resulting sign contrast: the actual and permuted vectors have the same norm and coordinate multiset, but their signed-Gram readouts differ sharply.

Every statement below is finite. A finite row count is not a uniformity quantifier, and a growth factor between two scales is not an asymptotic exponent.

2 The literal finite object

For an origin o and an even scale N , let

$$I_{o,N} = \{o, o + 1, \dots, o + N/2 - 1\}.$$

The protocol uses

$$o \in \{28001, 36001\}, \quad N \in \{4096, 8192\}, \quad H = 66,$$

and $Q \in \{24, 36, 54, 80\}$ with $s \in \{1, 2\}$. For a prime $p \in (Q, 2Q]$, define

$$B_{p,Q,s}(u, t) = \mathbf{1}_{u \neq t} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid t} p \frac{H^{2s}}{(H^2 + (u - t)^2)^s} \left(\mathbf{1}_{p \mid u - t} - \frac{1}{p - 1} \right). \quad (1)$$

The diagonal is deleted before the centered residue factor is applied. For a fixed sign law $e = (e_p)$ on the shell, put

$$C_e = \sum_{p \in (Q, 2Q]} e_p B_{p, Q, s}.$$

The four laws are all-plus, index-alternating, the sign of p modulo 4, and a half split in increasing prime order.

For a finite vector $v = (v_t)_{t \in I_{o, N}}$, define

$$E_e(v) = \|C_e v\|_2^2, \quad (2)$$

$$D_e(v) = \sum_t v_t^2 \|C_e e_t\|_2^2, \quad (3)$$

$$O_e(v) = E_e(v) - D_e(v). \quad (4)$$

When $D_e(v) > 0$, use $R_e(v) = E_e(v)/D_e(v)$. Thus $R_e < 1$ and $R_e > 1$ mean negative and positive off-diagonal Gram mass, respectively.

3 Source model and placement control

The source is the finite V59 comparison model

$$\Lambda(m) = \begin{cases} \log p, & m = p^k, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$b^{(2)}(t) = 2C_2 \mathbf{1}_{2|t} \prod_{\substack{p|t \\ p>2}} \frac{p-1}{p-2}, \quad \beta_o^{(2)}(t) = \Lambda(t+2) - b^{(2)}(t). \quad (5)$$

The executable certificate evaluates the product through 50000, with the inherited positive tail enclosure, and uses 100-digit Decimal logarithm centers with a rational 10^{-70} guard. The resulting midpoint vector is used in the float64 matrix replay.

The placement control is not selected after seeing the result. For both $M = 2048$ and $M = 4096$, $\gcd(5, M) = 1$, so π is a bijection. If P_π is the associated permutation matrix, then

$$P_\pi^T P_\pi = I, \quad \|P_\pi v\|_2 = \|v\|_2.$$

Thus the actual vector and $P_\pi v$ have identical coordinate multisets and norms. They need not have identical images under C_e : the literal divisibility masks and distance kernel are not assumed to commute with an arbitrary coordinate permutation.

4 Exact finite identities

Gram decomposition. The finite product expands as $C_e v = \sum_t v_t C_e e_t$. Therefore, by bilinearity,

$$E_e(v) = \sum_{t, t'} v_t v_{t'} \langle C_e e_t, C_e e_{t'} \rangle.$$

The terms with $t = t'$ are $D_e(v)$ and the remaining terms are

$$O_e(v) = \sum_{t \neq t'} v_t v_{t'} \langle C_e e_t, C_e e_{t'} \rangle.$$

Hence $E_e(v) = D_e(v) + O_e(v)$ exactly for every finite vector. No limiting interchange or arithmetic estimate occurs in this step; this is the finite Gram expansion in the usual matrix-analysis sense [1].

Placement identity. The permutation matrix identity above proves norm and multiset preservation. On the other hand,

$$E_e(P_\pi v) = v^T P_\pi^T C_e^T C_e P_\pi v.$$

Equality with $E_e(v)$ for all v would require a commutation property of the Gram operator that is not part of the model. The finite control therefore tests coordinate placement while fixing the source norm.

5 Certificate protocol

The producer writes canonical JSON containing all 32 actual rows and their placement-control rows. Each row stores source geometry, shell, matrix shape, the two component controls, E, D, O, R , and the outward ratio guard. A row is labelled negative when $R + 5 \cdot 10^{-8} < 1$, positive when $R - 5 \cdot 10^{-8} > 1$, and unresolved otherwise.

The independent checker does not import the producer. It rebuilds the source with separate factorization code, accumulates shell blocks in reverse order, recomputes actual and permuted metrics, and checks every stored value. It also replays the 64 scale pairs. The stress suite checks exact rational Gram algebra, source/sign mutations, deleted diagonals, permutation metadata, provenance hashes, and the fail-closed claim firewall. Normal and optimized executions must have empty standard error and byte-identical output.

6 Finite results

Table 1: Actual and norm-preserving placement-control censuses.

sign law	actual		permuted	
	$O < 0$	$O > 0$	$O < 0$	$O > 0$
all-plus	31	1	0	32
alternating index	25	7	30	2
mod-4 character	32	0	32	0
half split	32	0	28	4

No actual or permuted row is unresolved under the declared guard. The all-plus classification changes on 31/32 comparisons. Its maximum absolute ratio difference is 3.1669505870381451; the corresponding maxima for the alternating, mod-4, and half-split laws are 0.2512622344035258, 0.21527386205638832, and 0.1981450927371966.

The actual all-plus ratio ranges from 0.57333063185858035 to 1.0091831972566414. The component controls remain positive on all rows:

$$\min R_+(\Lambda(\cdot + 2)) = 1.3668932693626414, \quad \min R_+(b^{(2)}) = 3.0441001012913311.$$

This rules out a zero-energy component as an explanation for the finite sign contrast, but it does not create an arithmetic estimate.

6.1 Two-scale audit

Pairing rows at fixed origin, shell anchor, and exponent gives 64 law-level comparisons from 4096 to 8192. For all-plus, signs persist in 15/16 pairs and cross once. The energy growth factor and base-2 log-slope ranges are

$$1.9663131482417533 \leq E_{8192}/E_{4096} \leq 2.14326466572482,$$

$$0.97549309860589706 \leq \log_2(E_{8192}/E_{4096}) \leq 1.0998100153677246.$$

The analogous sign persistence counts are 15/16, 16/16, and 16/16 for alternating, mod-4, and half-split laws. These two-point slopes are reported as numerical observations only.

7 Exact local anchor

At $I = [28001, 28016]$, $Q = 4$, and $s = 1$, the shell is $\{5, 7\}$. The exact rational vector is

$$v_t = \mathbf{1}_{t+2 \text{ is prime}} - \mathbf{1}_{t \text{ is odd}}.$$

The Fraction calculation gives, after decimal display,

$$E = 582.0014930371829, \quad D = 649.9719408544561, \quad O = -67.97044781727324,$$

and verifies $E = D + O$ before display. Reduced numerator/denominator digests for all three values are stored in the certificate and independently recomputed. The anchor is a finite arithmetic sanity check, not a claim about twin-prime density.

8 Interpretation and claim boundary

The strongest finite positive result is the held-out source-native replay plus the norm-preserving placement control. The strongest obstruction is the actual/permutated all-plus contrast: identical source multiset and L^2 norm do not determine the sign of this coordinate-sensitive diagnostic on the released panel. This is a scoped obstruction to a source-norm-only interpretation.

The following remain open or unpaid:

- a source-uniform growing L^2 estimate for the actual residual;
- a canonical arithmetic sign law and placement-aware reassembly bound;
- an operator-norm estimate with the required physical normalization;
- the strict 1/400 endpoint loss, Route-B Gate B, and any twin-prime conclusion.

Accordingly,

```
ARITHMETIC_ADVANCE=NO,  FIXED_POWER_CREDIT=0,  FULL_GATE_B=OPEN,  TWIN_PRIME_RESULT=NONE.
```

The Session-named Route-A and Route-B evaluator files are absent from this checkout; the local Bridge-B checker is a fail-closed fallback and is not an official evaluator pass.

9 Reproducibility

The source, canonical certificate, independent checker, stress suite, proof package, and route ledger are in [papers/tpc-329-heldout-growing-source-native-audit](#). From the repository root, run the commands listed in the project README. The final machine-readable result is the certificate JSON in **results/**; the compiled manuscript is **paper/paper.pdf**.

References

- [1] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, 2 edition, 2013.